Total No. of Questions : 6]

SEAT No. :

P973

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APR-17/B.E./Insem.-24

B.E. (Mechanical)

FINITE ELEMENT ANALYSIS

(Semester - II)

Time : 1 Hour] *[Max. Marks : 30*

Instructions to the candidates:

1) *Draw suitable neat diagrams, wherever necessary.*
2) *Figures to the right indicate full marks.*
3) *Use of electronic pocket calculator is allowed.*
4) *Assume suitable data, if required.*

Q1) a) Write a difference between Weighted Residual and Weak Formulation Method. **[6]**

b) Explain advantages of FEA method over Finite Difference Method. **[4]**

OR

Q2) a) Write a note on Raleigh Ritz method for Element Formulation and its advantages over other elemental formulation methods. **[6]**

b) Explain step by step procedure for Panelty Approach and Elimination Approach. **[4]**

Q3) Determine the nodal displacements at node 2, stresses in each material and support reactions in the bar shown in Fig., due to applied force P = 400 × 103N and temperature rise of 30°C. Given : **[10]**

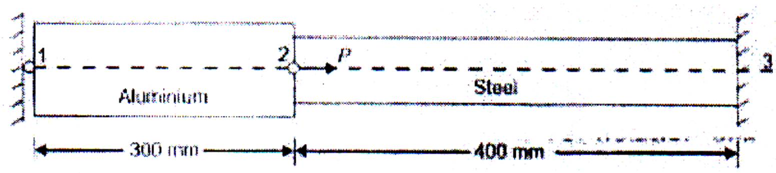
$A_1 = 2400\,\text{mm}^2 \;\; A_2 = 1200\,\text{mm}^2$

$l_1 = 300\,\text{mm} \;\; l_2 = 400\,\text{mm}$

$E_1 = 0.7 \times 10^5\,\text{N/mm}^2 \;\; E_2 = 2 \times 10^5\,\text{N/mm}^2$

and $\alpha_1 = 22 \times 10^{-6}/\text{C}° \;\; \alpha_2 = 12 \times 10^{-6}/\text{C}°$

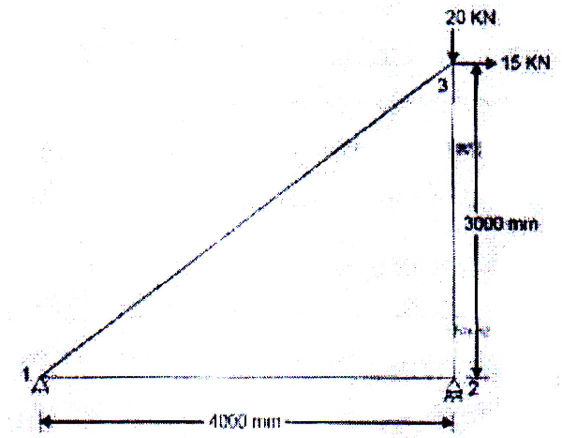
P.T.O.



$(A_1, l_1, E_1, \alpha_1)$ $(A_2, l_2, E_2, \alpha_2)$

OR

Q4) a) Obtain the forces in the plane truss shown in Fig. and determine the support reactions also. Use finite element method. Take E = 200 GPa and A = 2000 mm^2. **[6]**



b) Explain importance of Boundary conditions in FEA analysis and write what type of boundary conditions are applied for stress analysis. **[4]**

Q5) a) Explain LST element and write its stiffness matrix. **[6]**

b) Write a note on Pascal's Triangle for identification of 2D element interpolation function. **[4]**

OR

Q6) a) What is meant by Banded and Skyline Matrix methods and how these are used for reduction in memory required to simulation in FEA. **[6]**

b) Explain continuity requirements in FEA and write a note on C0 and C1 continuities with appropriate examples. **[4]**

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